

Education

University of Southampton (MEng Aerospace Engineering)

Sept 2024 – 2028 (est.)

Year 2 average **76.5%** (First Class)

Relevant modules:

- SESA2022 Aerodynamics: Wind tunnel testing of wings, data analysis and lab report
- SESA2029 CFD/FEA: Aerodynamic analysis of a wing in CFD
- SESA2027 Aerospace Mechanics and Control: development of a model and PID controller to dampen the longitudinal short-period mode of a Lockheed Jetstar.

Projects and Experience

StarWorks 2.5kN rocket-powered lander (Spaceflight Society project)

April 2025 – Present

Structures Team Leader

- **Team Leadership:** Lead a team of 10 engineers, delegating tasks, tracking progress and coordinating design reviews. Delivered the Preliminary Design Review (PDR) on schedule while mentoring new members in ANSYS and Onshape.
- **Cross-Team Coordination:** Liaised with propulsion, avionics and control teams to define dimensional/interface requirements, resolving component over-sizing and placement conflict issues.
- **Landing Dynamics Modelling:** Developed a Python landing dynamics model to size landing legs and energy absorbers for a 20 m landing.
- **Structural Verification:** Designed landing legs in Onshape and verified structural integrity using ANSYS Mechanical FEA.

Multi-rotor UAV University group project

Jan 2026 – May 2026

Coding and Integrated Systems with Arduino Nano

- **Gimbal Control:** Coded an open-loop stabilization scheme for a mode-switching 3-axis gimbal, utilising an angular-velocity damping term to improve response.
- **Sensor Fusion:** Fused accelerometer data via complementary filtering for pitch/roll, with yaw bias removed by measuring startup drift and subtracting it each loop.
- **Signal Filtering:** Applied first- and second-order filters to MPU6050 data to remove motor vibration and steady-state noise.
- **Telemetry and Validation:** Logged flight telemetry to an SD card and reconstructed the flight path in Python for validation against motion capture measurements.
- **Teamwork:** Worked alongside team members, assembling laser cut parts, wiring and soldering electronics, whilst independently owning gimbal design and coordinating electronics placement with the team.

Fixed-Wing Camera UAV

June 2026 – Present

Personal Project - 3D-printed Fixed-wing UAV

- **Wing Design:** Designed and optimised a low-Reynolds-number wing in XFLR5 around the SD7037 airfoil, balancing cruise-efficiency, low-speed performance and handling.
- **CFD Validation:** Validated XFLR5 predictions using Ansys Fluent with the Transition SST model and inflation-layer meshing to evaluate drag, lift and stall behaviour.
- **Systems Integration:** Sized the electrical power system, selected ESCs and optimised component layout for centre-of-gravity and wiring constraints.
- **Propulsion Optimisation:** Optimised propeller/motor combinations using Python interpolation of APC data to maximise endurance while satisfying thrust requirements.

Volunteering and Other Work

Scouting Association

Sept 2022 – July 2024

Assistant Scout Leader

- Led and supervised groups, encouraging teamwork and problem solving through technical and outdoor activities, whilst maintaining safety

Local Pub

Sept 2023 – Jan 2024

Bar/Waiting Staff

Skills

Software: Ansys (Fluent, Mechanical, Mechanical APDL), XFLR5, SOLIDWORKS, Onshape, LaTeX, OpenVSP

Coding: Python, C#, Arduino IDE (Arduino Nano + ESP32)

Engineering: CFD, FEA, Flight dynamics, Aerodynamics, Embedded systems, Wind tunnel testing, Data analysis

Manufacturing: 3D printing, Soldering, Milling, Lathe

Interests and Hobbies

Personal aircraft design projects, RC aircraft, self-taught bass and guitar, Lego modelling, Duke of Edinburgh Gold Award.